



2023 YEAR 12 TASK 4

Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes
- Working time – 180 Minutes
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total Marks:
100**

Section I – 10 marks

- Attempt Questions 1–10
- Allow about 20 minutes for this section

Section II – 90 marks

- Attempt Questions 11 – 14
- Allow about 160 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Use the multiple-choice answer sheet for Questions 1–10.

- 1 In an Argand diagram the points $A(-3, 2)$ and $B(5, -4)$ lie at opposite ends of a diameter of a circle. What is the equation of the circle?

A. $|z - 1 + i| = 5$

B. $|z + 1 - i| = 5$

C. $|z - 1 + i| = 10$

D. $|z + 1 - i| = 10$

- 2 What is the size of the acute angle θ between the vectors $\underline{a} = 2\underline{i} - \underline{j} - \underline{k}$ and $\underline{b} = 2\underline{i} - 2\underline{k}$?

A. $\theta = \frac{\pi}{6}$

B. $\theta = \frac{\pi}{5}$

C. $\theta = \frac{\pi}{4}$

D. $\theta = \frac{\pi}{3}$

3 Which of the following is an expression for $\int \frac{1}{x^2 - \sqrt{3}x + 1} dx$?

A. $\tan^{-1}(x - \sqrt{3}) + c$

B. $2\tan^{-1}(x - \sqrt{3}) + c$

C. $\tan^{-1}(2x - \sqrt{3}) + c$

D. $2\tan^{-1}(2x - \sqrt{3}) + c$

4 The amount of apples, bananas and oranges sold by a fruit seller over a year is shown in the table below

Fruit	Amount Sold (tonnes)	Profit (\$/tonne)
Apples	25	530
Bananas	55	380
Oranges	10	410

Let $a = \begin{bmatrix} 25 \\ 55 \\ 10 \end{bmatrix}$, $b = \begin{bmatrix} 530 \\ 380 \\ 410 \end{bmatrix}$ and $c = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Which of the following expressions calculates the average profit in dollars per tonne of fruit sold over the year?

A. $\frac{a \cdot c}{b \cdot c}$

B. $\frac{b \cdot c}{a \cdot c}$

C. $\frac{a \cdot b}{b \cdot c}$

D. $\frac{a \cdot b}{a \cdot c}$

5 Which of the following expressions is equivalent to $\int \ln(x^2 + 1) \, dx$?

A. $x \ln(x^2 + 1) - 2x + 2 \tan^{-1} x + c$

B. $x \ln(x^2 + 1) - 2 \ln(x^2 + 1) + c$

C. $\ln(x^2 + 1) - 2x + 2 \tan^{-1} x + c$

D. $\ln(x^2 + 1) - x \ln(x^2 + 1) + c$

6 Consider the complex, non-real cube roots of unity ω and ω^2 .

What is the value of $(1 - \omega + \omega^2)(1 + \omega - \omega^2)$?

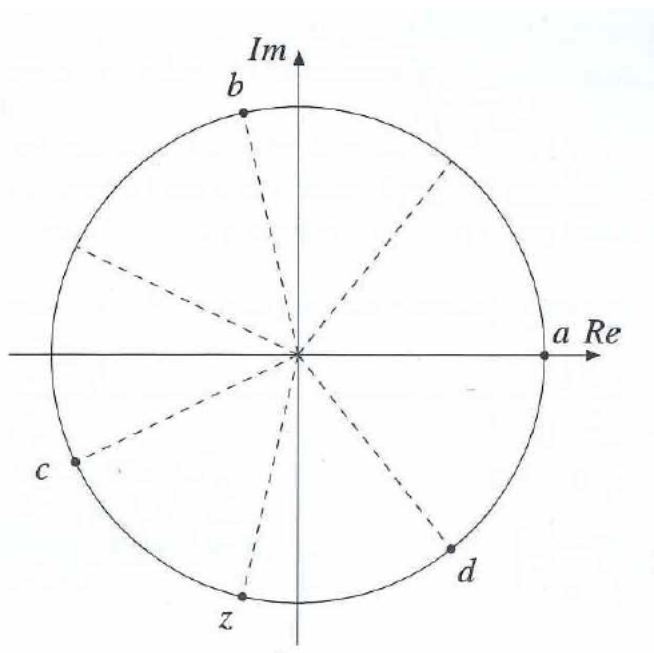
A. 0

B. 1

C. 2

D. 4

- 7 The complex numbers a, b, c, d and z are solutions to $z^7 = 1$ as shown in the Argand diagram below.



Which of the following is a cube root of z ?

- A. a
- B. b
- C. c
- D. d

- 8 A sequence of complex numbers $z_1, z_2, z_3, z_4, \dots$ is given by the rule $z_1 = Z$ and $z_{n+1} = c\bar{z}_n + c - 1$ for $n \in \mathbb{Z}^+$ where c is a complex number with modulus 1. What is the value of z_3 ?

- A. $z_3 = Z$
- B. $z_3 = -2 + Z$
- C. $z_3 = 2c + Z$
- D. $z_3 = -2 + 2c + Z$

- 9 Consider a line that passes through the point $(5, 2, 1)$ and is parallel to the $x - y$ plane and the $x - z$ plane.

Which of the following is the vector equation of the line?

A. $\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

B. $\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

D. $\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

10 The function $F(x)$ is the primitive of $f(x)$, that is $F'(x) = f(x)$. Which of the following is true?

A. $\int \left(\frac{d}{dx} \int_a^b f(x) dx \right) dx = f(b) - f(a)$

B. $\int_a^b \left(\frac{d}{dx} \int f(x) dx \right) dx = f(b) - f(a)$

C. $\frac{d}{dx} \int \left(\int_a^b f(x) dx \right) dx = F(b) - F(a)$

D. $\frac{d}{dx} \int_a^b \left(\int f(x) dx \right) dx = F(b) - F(a)$

Section II

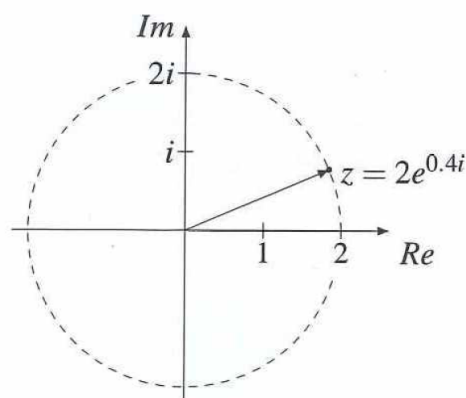
90 marks

Attempt Questions 11 – 16

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Please begin a new Writing Booklet.

- (a) Express $2\sqrt{2}e^{-\frac{3\pi}{4}i}$ in the form $x + iy$. 2
- (b) Consider the two points $A(2, 2, 2)$ and $B(2, -2, 2)$.
- (i) Find $\overrightarrow{AB}$ 1
- (ii) Find $|\overrightarrow{AB}|$ 1
- (iii) Find $\angle AOB$. Give your answer to the nearest degree. 2
- (c) Consider the complex number $z = 2e^{0.4i}$, as sketched below. 3



Copy and clearly label this Argand diagram, and sketch the four points represented by z , $\bar{z}$, $-\bar{z}$ and $z - \bar{z}$.

(d) Use the substitution $t = \tan \frac{x}{2}$ to find $\int \frac{1}{1 + \sin x} dx$. **3**

(e) Show for any complex numbers z and w , that $\overline{zw} = \bar{z} \times \bar{w}$. **3**

End of Question 11

Question 12 (15 marks) Please begin a new Writing Booklet.

(a) Consider the two lines

$$L_1 : \begin{bmatrix} 7 \\ -1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix} \text{ and } L_2 : \begin{bmatrix} -1 \\ 2 \\ -6 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}.$$

(i) Find the values of k given $B(9, k, 24)$ lies on L_2 . **2**

(ii) Find the point of intersection of L_1 and L_2 . **3**

(b) Find $\int_0^{\frac{1}{\sqrt{2}}} \frac{x^3}{\sqrt{1-x^2}} dx$ **4**

(c) Solve $z^2 + (7-i)z + 16 + 4i = 0$ **3**

(d) Find $\int \frac{x^3 - 2}{x^3 - x} dx$. **3**

End of Question 12

Question 13 (15 marks) Please begin a new Writing Booklet.

- (a) On an Argand diagram shade the region containing all points representing the complex numbers z that satisfy both $|z - 2i| \leq 2$ and $\frac{\pi}{6} \leq \arg z \leq \frac{\pi}{4}$. **2**
- (b) The graph of a polynomial function $f(x) = (x + 3)(x - 2)(x^2 + bx + c)$ has a y intercept of -6 and passes through $(1, -4)$.
- (i) Find the two complex roots of the equation $f(x) = 0$. **2**
- (ii) Express these two complex roots in the form $r(\cos \theta + i \sin \theta)$. **2**
- (iii) Plot all four solutions to $f(x) = 0$ on an Argand diagram. **2**
- (iv) Write down the name of the quadrilateral formed by these four points. **1**
- (c) Consider the sphere with vector equation $\left| \underline{r} - \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix} \right| = 3$.
- (i) Show that the point $(5, -10, 3)$ lies on the sphere. **1**
- (ii) Find the point on the sphere farthest from the origin. **2**
- (d) Prove by Mathematical induction that $3n^2 - 3n \leq 2^n - 1$ for $n \geq 7$. **3**

End of Question 13

Question 14 (15 marks) Please begin a new Writing Booklet.

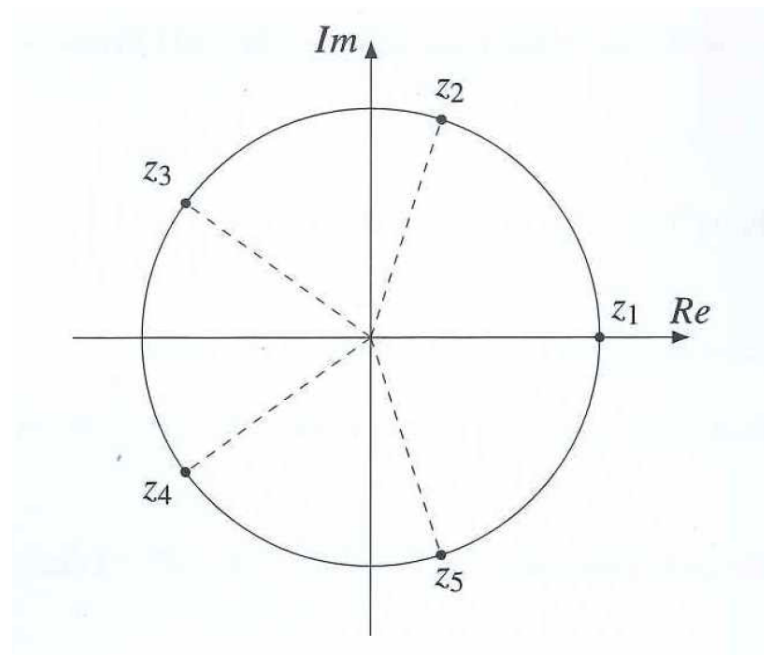
(a) The number z is a fifth root of unity where $z \neq 1$, that is $z^5 = 1$

(i) Show that $(z + z^{-1})^2 + (z + z^{-1}) - 1 = 0$. 2

(ii) If $z = e^{i\theta}$, show $\cos \theta = \frac{z + z^{-1}}{2}$. 1

(iii) Hence show $\cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}$ 2

(iv) Consider the five fifth roots of unity z_1, z_2, z_3, z_4 and z_5 as shown in the diagram 3
below.



Show that $\left| \frac{z_3 - z_1}{z_2 - z_1} \right| = \frac{1 + \sqrt{5}}{2}$.

Question 14 continues on page 13

Question 14 (continued)

(b) The position of an object after t seconds is given by the vector equation

$$\underline{r} = \cos \frac{\pi t}{4} \underline{i} + \left(\cos \frac{\pi t}{4} + \sin \frac{\pi t}{4} \right) \underline{j} + \sin \frac{\pi t}{4} \underline{k}.$$

(i) What is the position of the object after 3 seconds? **1**

(ii) Find the vector equation of the tangent to the path taken by the object after 3 seconds. **3**

(c) Find $\int_0^9 \frac{1}{\sqrt{1+\sqrt{x}}} dx$ **3**

End of Question 14

Question 15 (15 marks) Please begin a new Writing Booklet.

- (a) Consider the function $f(x) = e^{-x} \cos x$, with domain $x \in \left[0, \frac{3\pi}{2}\right]$. Show that the ratio of the area above the x -axis to the area below the x -axis is **4**

$$\frac{e^{\pi} \left(e^{\frac{\pi}{2}} + 1 \right)}{e^{\pi} + 1}$$

- (b) A series is defined by $T_n = 6T_{n-1} - 9T_{n-2}$ for $n \geq 2$ where $T_0 = 1$ and $T_1 = 6$. **4**

Use Mathematical induction to prove $T_n = 3^n + n \times 3^n$.

(c) Let $I_n = \int \frac{\cos nx}{\sin x} dx$.

- (i) Show that $\cos((n-2)x) - \cos nx = 2 \sin((n-1)x) \sin x$. **2**

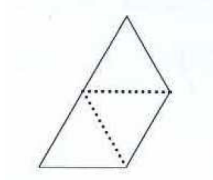
- (ii) Show that $I_n - I_{n-2} = \frac{2 \cos((n-1)x)}{n-1} + c$ **2**

- (iii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{3}} \frac{\cos 2x - \cos 6x}{\sin x} dx$ **3**

End of Question 15

Question 16 (15 marks) Please begin a new Writing Booklet.

- (a) Consider the tile below consisting of three equilateral triangles of side length 1 unit. **3**



Prove the following result for positive integers n , using Mathematical induction:

An equilateral triangle of side length 2^n units may be covered by the tiles shown above (in any orientation) such that a single equilateral triangle of side length 1 unit is left over at one of the vertices. The tiles may not overlap.

- (b) The limiting sum of series is given by $S = \frac{a}{1-r}$ where a is the first term of the series and r is the common ratio between consecutive terms. It only exists when $|r| \leq 1$.

- (i) Show that the limiting sum S of the series $1 + \frac{1}{2}e^{i\theta} + \frac{1}{4}e^{2i\theta} + \frac{1}{8}e^{3i\theta} + \dots$ is given by **2**
$$S = \frac{2}{2 - e^{i\theta}}.$$

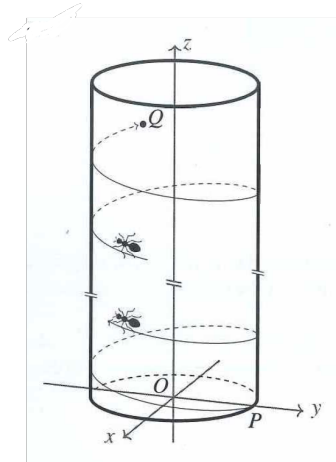
- (ii) Hence show that $\frac{1}{2} \sin \theta + \frac{1}{4} \sin 2\theta + \frac{1}{8} \sin 3\theta + \dots = \frac{2 \sin \theta}{5 - 4 \cos \theta}$. **2**

- (iii) Show that there exists no real values of θ such that S is purely imaginary. **1**

Question 16 continues on page 16

Question 16 (continued)

- (c) An ant follows a spiral path up a cylindrical column from $P(0, 7, 0)$ to the point Q as shown in the diagram below.



The ant's position r in centimetres after t seconds is given by the vector equation below.

$$r(t) = 7 \sin \frac{\pi t}{32} i + 7 \cos \frac{\pi t}{32} j + \frac{t}{15} k$$

- | | |
|---|----------|
| (i) Find the coordinates of the point Q if $ \vec{OQ} = 25$. | 3 |
| (ii) How many times has the ant crossed the line $x = 7$ on its journey to Q ? | 2 |
| (iii) The ant crawls back from Q to P along the shortest path possible. How far did it crawl on this leg of its journey? Give your answer correct to one decimal place. | 2 |

End of Exam

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

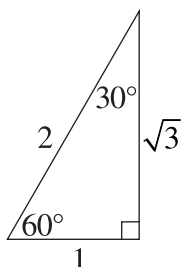
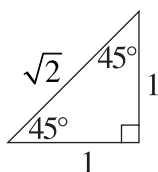
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

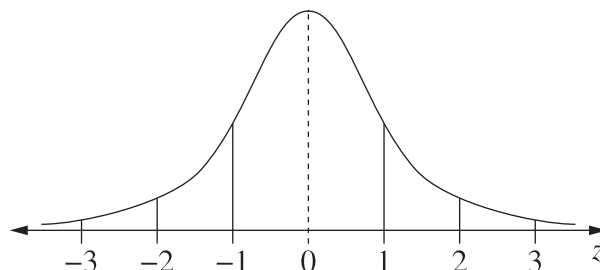
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

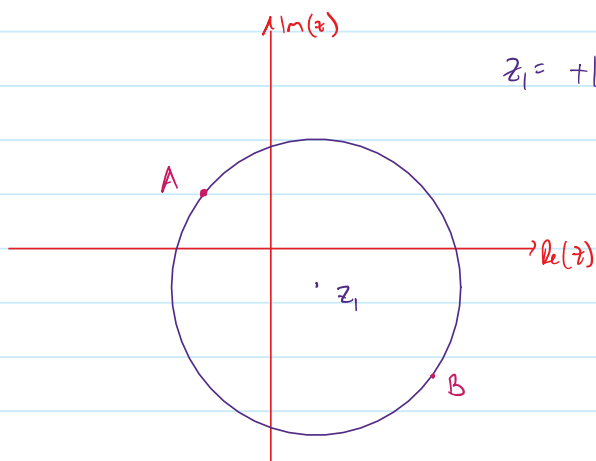
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12 Ext 2 2023 Task 4 - RDS Solutions MC

Monday, 31 July 2023

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Q1



$$z_1 = 1 - i$$

$\therefore$ Centre @ $z = 1 - i$

$$\text{Radius} = \frac{1}{2} (\sqrt{8^2 + 6^2})$$

$$= 5$$

$$\therefore |z - (1 - i)| = 5 \quad \therefore \text{A}$$

Q2

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$|\underline{a}| = \sqrt{6} \quad |\underline{b}| = 2\sqrt{2}$$

$$\underline{a} \cdot \underline{b} = 4 + 0 + 2 = 6$$

$$\therefore \cos \theta = \frac{6}{2\sqrt{2} \cdot \sqrt{6}}$$

$$= \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{6} \quad \therefore \text{A}$$

Q3

$$\int \frac{1}{x^2 - \sqrt{3}x + 1} dx$$

$$= \int \frac{1}{\frac{1}{4}((2x - \sqrt{3})^2 + 1)} dx$$

$$= 4 \tan^{-1} \left(\frac{2x - \sqrt{3}}{1} \right) \div 2 + C$$

$$x^2 - \sqrt{3}x + 1$$

$$= x^2 - \sqrt{3}x + \left(\frac{\sqrt{3}}{2}\right)^2 + 1 - \left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \left(x - \frac{\sqrt{3}}{2}\right)^2 + \frac{1}{4}$$

$$= \frac{1}{4} (2x - \sqrt{3})^2 + \frac{1}{4}$$

$$= \frac{1}{4} [(2x - \sqrt{3})^2 + 1]$$

$$= 2 \tan^{-1}(2x - \sqrt{3}) + C \quad \therefore \textcircled{D}$$

Q4 $\bar{x} = \frac{\text{Amount} \times \text{Profit}}{\text{total fruit sold}}$

$$\text{Amount} \times \text{Profit} = \underline{a} - \underline{b}$$

$$\text{total fruit} = \underline{a} \cdot \underline{c}$$

$$\therefore \bar{x} = \frac{\underline{a} - \underline{b}}{\underline{a} \cdot \underline{c}} \quad \therefore \textcircled{D}$$

Q5 $\int \ln(x^2+1) dx \quad u = \ln(x^2+1) \quad v = x$

$$\frac{du}{dx} = \frac{2x}{x^2+1} \quad \frac{dv}{dx} = 1$$

$$\therefore I = x \ln(x^2+1) - \int \frac{2x^2}{x^2+1} dx$$

$$= x \ln(x^2+1) - 2 \int \frac{x^2+1}{x^2+1} - \frac{1}{x^2+1} dx$$

$$= x \ln(x^2+1) - 2(x - \tan^{-1} x) + C$$

$$\therefore \textcircled{A}$$

Q6 $(1-\omega+\omega^2)(1+\omega-\omega^2)$

$$1+\omega+\omega^2 = 0$$

$$\therefore 1+\omega^2 = -\omega$$

$$1+\omega = -\omega^2$$

$$= (-2\omega)(-2\omega^2)$$

$$= 4\omega^3$$

$$\omega^3 = 1$$

$$= 4 \quad \therefore \textcircled{D}$$

Q7 $\text{Arg}(a) = 0 \times 3 = 0$

$\text{Arg}(b) = \frac{4\pi}{7} \times 3 = \frac{12\pi}{7}$

$\text{Arg}(c) = \frac{8\pi}{7} \times 3 = \frac{24\pi}{7} = \frac{10\pi}{7}$

$\text{Arg}(z) = \frac{10\pi}{7}$

$\therefore \textcircled{C}$

$\text{Arg}(d) = \frac{12\pi}{7} \times 3 = \frac{36\pi}{7} = \frac{12\pi}{7}$

Q8 $z_{n+1} = c\bar{z}_n + c - 1$

$z_2 = c\bar{z} + c - 1$

$z_3 = c \times \left[c\bar{z} + c - 1 \right] + c - 1$

$= c \times \left[\bar{c}z + \bar{c} - 1 \right] + c - 1$

$= c \times \bar{c}z + \cancel{c\bar{c}} - \cancel{c} + \cancel{c} - \cancel{1}$

$c \times \bar{c} = 1$

$= z \therefore \textcircled{A}$

Q9 To be parallel to x-y plane & the z-z plane the line must be parallel to the z axis.

$\therefore \text{line} : \begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \therefore \textcircled{C}$

Q10 $\frac{d}{dx} \int_a^b f(x) dx = 0$ ($\int_a^b f(x) dx$ gives a constant)

$\therefore$ Not $\textcircled{A}$ or $\textcircled{D}$

$\frac{d}{dx} \int f(x) dx = \frac{d}{dx} (f(x) + c)$
 $= f(x)$

$$\frac{d}{dx} \int_a^x f(x) dx = f(x)$$

$$\therefore \int_a^b f(x) dx = f(b) - f(a) \quad \therefore \text{Not } \textcircled{B}$$

$$\int_a^b f(x) dx = f(b) - f(a)$$

$$\int f(b) - f(a) dx = [f(b) - f(a)]x + C$$

$$\frac{d}{dx} [f(b) - f(a)]x = f(b) - f(a) \quad \therefore \textcircled{C}$$

Answers.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
A	A	D	D	A	D	C	A	C	C

12 Ext 2 2023 Task 4 - RDS Solutions Q11

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$$\begin{aligned}
 (a) \quad 2\sqrt{2}e^{-\frac{3\pi}{4}i} &= 2\sqrt{2} \left(\cos \frac{-3\pi}{4} + i \sin \frac{-3\pi}{4} \right) \\
 &= 2\sqrt{2} \left(-\frac{1}{\sqrt{2}} - i \frac{1}{\sqrt{2}} \right) \\
 &= -2 - 2i
 \end{aligned}$$

$$(b) \quad A(2, 2, 2), \quad B(2, -2, 2)$$

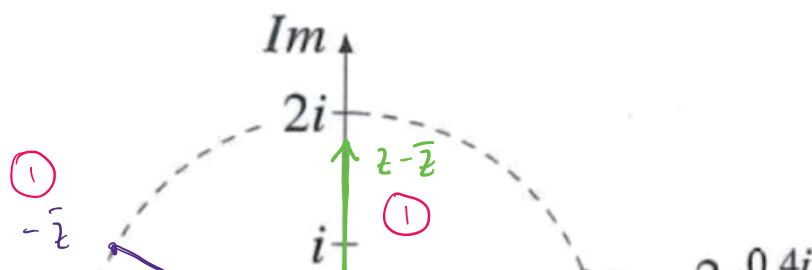
$$\begin{aligned}
 (i) \quad \vec{AB} &= \vec{OB} - \vec{OA} \\
 &= -4\mathbf{j}
 \end{aligned}$$

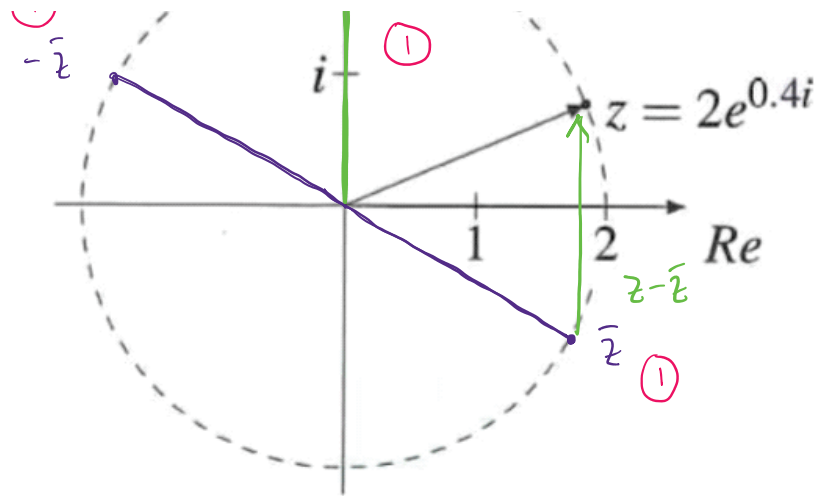
$$(ii) \quad |\vec{AB}| = 4$$

$$\begin{aligned}
 (iii) \quad \cos \theta &= \frac{\vec{OA} \cdot \vec{OB}}{|\vec{OA}| |\vec{OB}|} \\
 &= \frac{4}{2\sqrt{3} \times 2\sqrt{3}} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \theta &= \cos^{-1} \left(\frac{1}{3} \right) \\
 &\approx 71^\circ \quad (\text{Nearest degree})
 \end{aligned}$$

(c)





(d) $t = \tan \frac{x}{2}$

$\therefore x = 2 \tan^{-1} t$

$\frac{dx}{dt} = 2 \left(\frac{1}{1+t^2} \right)$

$\therefore dx = \frac{2dt}{1+t^2}$

$\int \frac{1}{1+\sin x} dx = \int \frac{1}{1 + \frac{2t}{1+t^2}} \times \frac{2dt}{1+t^2}$

$= \int \frac{2}{t^2 + 2t + 1} dt$

$= \int \frac{2}{(t+1)^2} dt$

$= -2(t+1)^{-1} + C$

$= -2 \left[\tan \left(\frac{x}{2} \right) + 1 \right]^{-1} + C$

(e) $\overline{z\omega} = \bar{z} \times \bar{\omega}$

let $z = a+ib$, $\omega = x+iy$

let $z = a+ib$, $w = x+iy$ ①

$$\overline{zw} = \overline{(a+ib)(x+iy)}$$

$$= \overline{(ax+ibx+iaiy-by)}$$

$$= \overline{(ax-by) + i(bx+ay)}$$

$$= (ax-by) - i(bx+ay)$$

$$\bar{z} \times \bar{w} = (a-ib)(x-iy)$$

$$= ax - iay - ibx - by$$

$$= (ax-by) - i(ay+bx)$$

$$= \overline{zw}$$

As required.

12 Ext 2 2023 Task 4 - RDS Solutions Q12

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$$(a) (i) L_2: \begin{bmatrix} -1 \\ 2 \\ -6 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix} = \begin{bmatrix} 9 \\ k \\ 24 \end{bmatrix}$$

$$\text{Comparing } x: -1 + 2\mu = 9$$

$$\mu = 5 \quad (1)$$

$$\text{Comparing } y: 2 - 3\mu = k$$

$$2 - 15 = k$$

$$k = -13 \quad (1)$$

$$(ii) L_1: x = 7 + \lambda \quad L_2: x = -1 + 2\mu$$

$$y = -1 + 3\lambda \quad y = 2 - 3\mu$$

$$z = 2 - 5\lambda \quad z = -6 + 6\mu$$

Solve x/y

$$7 + \lambda = -1 + 2\mu \quad -1 + 3\lambda = 2 - 3\mu$$

$$\lambda - 2\mu = -8 \quad (1) \quad 3\lambda + 3\mu = 3$$

$$\lambda + \mu = 1 \quad (2)$$

(1) - (2)

$$-3\mu = -9$$

$$\mu = 3$$

$$\therefore \lambda = -2$$

Sub into z equations

$$\text{When } \lambda = -2 \quad \mu = 3$$

$$L_1: z = 12 \quad L_2: z = 12 \quad \therefore \text{Intersect.}$$

$\therefore$ Point of Intersection (sub $\lambda = -2$ into L_1)

$$P \begin{pmatrix} 5 \\ -7 \\ 12 \end{pmatrix}$$

$$(b) I = \int_0^{\frac{1}{\sqrt{2}}} \frac{x^3}{\sqrt{1-x^2}} dx \quad \text{let } x = \sin \theta \quad x = \frac{1}{\sqrt{2}}, \theta = \frac{\pi}{4}$$

$$\frac{dx}{d\theta} = \cos \theta \quad x = 0, \theta = 0 \quad (1)$$

$$\therefore I = \int_0^{\frac{\pi}{4}} \frac{\sin^3 \theta}{\sqrt{1-\sin^2 \theta}} \cdot d\theta \cdot \cos \theta \quad (1)$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin^3 \theta}{\cancel{\sin \theta}} \cdot \cancel{\sin \theta} \cdot d\theta$$

$$= \int_0^{\frac{\pi}{4}} \sin \theta (1 - \cos^2 \theta) d\theta$$

$$\sim \sim \sim \frac{\pi}{4}$$

$$\begin{aligned}
 &= \int_0^{\frac{\pi}{4}} \sin(\theta) \cos^2(\theta) d\theta \\
 &= \left[-\cos\theta + \frac{\cos^3\theta}{3} \right]_0^{\frac{\pi}{4}} \quad (1) \\
 &= \left(-\frac{1}{\sqrt{2}} + \frac{1}{3} \times \frac{1}{2\sqrt{2}} \right) - \left(-1 + \frac{1}{3} \right) \\
 &= \frac{8 - 5\sqrt{2}}{12} \quad (1)
 \end{aligned}$$

(c) $z^2 + (7-i)z + 16+4i = 0$

$$\begin{aligned}
 \Delta &= (7-i)^2 - 4(16+4i) \\
 &= 49 - 14i + 1 - 64 - 16i \\
 &= -16 - 30i \quad (1)
 \end{aligned}$$

let $(x+iy)^2 = -16-30i$ $x, y \in \mathbb{R}$

$$\begin{aligned}
 \therefore x^2 - y^2 &= -16 \quad (2) & 2xy &= -30 \\
 y &= \frac{-15}{x} \quad \dots (1)
 \end{aligned}$$

sub (1) into (2)

$$x^2 - \frac{225}{x^2} = -16$$

$$x^4 + 16x^2 - 225 = 0$$

$$(x^2+25)(x^2-9) = 0 \quad x \in \mathbb{R}$$

$$\therefore x = \pm 3, \quad y = \mp 5$$

$$\therefore \sqrt{\Delta} = \pm (3-5i) \quad (1)$$

$$\therefore z = \frac{-(7-i) \pm (3-5i)}{2}$$

$$\begin{aligned}
 z &= \frac{-10+6i}{2}, \quad \frac{-4-4i}{2} \\
 &= -5+3i, \quad -2-2i \quad (1)
 \end{aligned}$$

(d) $\int \frac{x^3-2}{x^3-x} dx = \int \frac{x^3-2}{x(x^2-1)} dx$

$$= \int \frac{x^3-2}{x(x-1)(x+1)} dx$$

let $\frac{x^3-2}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1} + \frac{D}{x-1}$

(1) Attempting partial fractions

$$x^3 - 2 = Ax(x-1)(x+1) + B(x+1)(x-1) + Cx(x-1) + D_1(x+1)$$

When $x=0$ $-2 = 0 - B + 0 + 0$
 $B = 2$

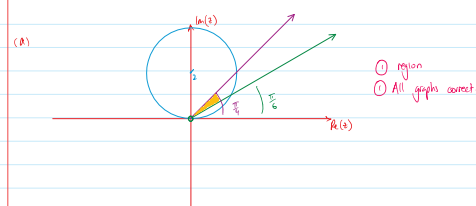
$x=1$ $-1 = 0 + 0 + 0 + 2D$
 $D = -\frac{1}{2}$

$x=-1$ $-3 = 0 + 0 + 2C + 0$
 $C = -\frac{3}{2}$

$x=2$ $6 = 6A + 3B + 2C + 6D$
 $6A = \cancel{6} - \cancel{6} + 3 + 3$
 $= 1$

$$\therefore \frac{x^3 - 2}{x^3 - x} = 1 + \frac{2}{x} - \frac{3}{2(x+1)} - \frac{1}{2(x-1)} \quad (1)$$

$$\therefore \int \left(1 + \frac{2}{x} - \frac{3}{2(x+1)} - \frac{1}{2(x-1)} \right) dx = x + 2\ln|x| - \frac{3}{2}\ln|x+1| - \frac{1}{2}\ln|x-1| + C \quad (1)$$



(b) $f(x) = (x+3)(x-2)(x^2+bx+c)$

(i) $f(0) = -6$, $f(1) = -4$

$\therefore -6 = -6c$
 $\therefore c = 1$

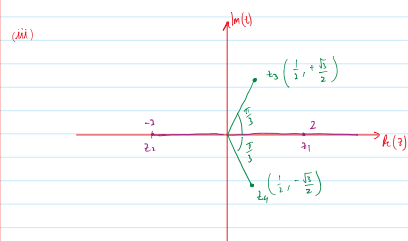
$-4 = -4(2+b)$
 $2+b = 1$
 $b = -1$ ①

$\therefore f(x) = (x+3)(x-2)(x^2-x+1)$

Solving $x^2-x+1 = 0$

$x = \frac{1 \pm \sqrt{1-4}}{2}$
 $= \frac{1 \pm \sqrt{3}i}{2}$ ①

(ii) $\left| \frac{1+\sqrt{3}i}{2} \right| = 1$
 $\arg\left(\frac{1+\sqrt{3}i}{2}\right) = \frac{\pi}{3}$ $\arg\left(\frac{1-\sqrt{3}i}{2}\right) = -\frac{\pi}{3}$ ①
 $\therefore$ roots are $1\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ & $1\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)$ ①



$z_1 = 1$
 $z_2 = 1$
 $z_3 = \cos\frac{\pi}{3}$
 $z_4 = \cos\frac{\pi}{3}$ ①
 Where $\cos\theta = \cos\theta + i\sin\theta$

① All parts correct
① All information added.

(iv) Kite $\rightarrow$ Diagonal are perpendicular
 $\rightarrow$ Adjacent sides equal ①

(c) (i) $\begin{bmatrix} 5-3 \\ -0+12 \\ 3-4 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}$

$\left| \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix} \right| = \sqrt{9} = 3$ ①

$\therefore$ Satisfies the equation
 $\therefore$ lies on the sphere.

(ii) let the point on the sphere with maximum distance from the origin be $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$

As this goes through the centre of the sphere

$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = k \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix}$ ①
 As the vector will be parallel to the position vector of the centre.

We also know that our vector lies on the sphere

$\therefore \frac{a-3}{b+12} = 3$ ②
 $\frac{c-4}{-1} = 3$

$\rightarrow \text{OR } \hat{r}_0 = \text{centre} = \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix}$

$\hat{r}_0 = \frac{1}{\sqrt{3^2 + (-12)^2 + 4^2}}$
 $= \frac{1}{13} \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix}$ ✓

$\therefore$ Further point is scalar multiple of centre position vector, $(+3)$ to negative

$\therefore \frac{1}{13} \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix} \times (3+13)$

$$\begin{vmatrix} 0+16 & - & - \\ C-4 & & \end{vmatrix}$$

Substituting ① into ②

$$\begin{vmatrix} 3k-3 \\ -12k+12 \\ 4k-4 \end{vmatrix} = 3$$

$$9(k-1)^2 + 144(k-1)^2 + 16(k-1)^2 = 9$$

$$169(k-1)^2 = 9$$

$$(k-1)^2 = \frac{9}{169}$$

$$k-1 = \frac{3}{13}$$

$$k = \frac{16}{13}$$

(k > 0 is local maximum)

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \frac{16}{13} \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix}$$

vector, (+) to magnitude

$$\therefore \frac{1}{13} \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix} \times (3+13)$$

$$= \frac{16}{13} \begin{bmatrix} 3 \\ -12 \\ 4 \end{bmatrix}$$

(d) Prove by Mathematical induction that $3n^2 - 3n \leq 2^n - 1$ for $n \geq 7$.

3

STEP ①: Prove true for $n=7$:

$n=7$
 LHS = $3(7)^2 - 3(7) = 147 - 21 = 126$
 RHS = $2^7 - 1 = 128 - 1 = 127$
 $126 \leq 127$
 $\therefore$ True for $n=7$.

STEP ②: Assume true for $n=k$

i.e. Assume $3k^2 - 3k \leq 2^k - 1$
 $3k^2 \leq 2^k + 3k - 1$
 $k^2 \leq \frac{2^k}{3} + k - \frac{1}{3}$

STEP ③: Prove true for $n=k+1$

MP: $3(k+1)^2 - 3(k+1) \leq 2^{k+1} - 1$

LHS = $3(k^2 + 2k + 1) - 3k - 3$
 $= 3k^2 + 6k + 3 - 3k - 3$
 $= 3k^2 + 3k$
 $= 3k^2 - 3k + 6k$
 $\leq 2^k - 1 + 6k$, from assumption.
 $\leq 2^{k+1} - 1$
 $\hookrightarrow$ since: $\begin{bmatrix} 2^{k+1} - 1 \geq 2^k - 1 + 6k. \\ 2^{k+1} \geq 2^k + 6k, \text{ true for } k \geq 7 \end{bmatrix}$

$\therefore$ True for $n=k+1$

① Use of assumption correctly.

① Proof correct.

(CONCLUSION:

$\therefore$ Proven true by M.I.

OR

$$2^k \geq 3k^2 - 3k + 1$$

③ RTP: $3(k+1)^2 - 3(k+1) \leq 2^{k+1} - 1$
 $3(k^2 + 2k + 1) - 3k - 3 \leq 2^{k+1} - 1$
 $3k^2 + 6k + 3 - 3k - 3 \leq 2^{k+1} - 1$
 $3k^2 + 3k + 1 \leq 2^{k+1}$

LHS = 2^{k+1}
 $= 2(2^k)$
 $\geq 2(3k^2 - 3k + 1)$
 $\geq 3k^2 + 3k + 1$

$\hookrightarrow$ since $2(3k^2 - 3k + 1) \geq 3k^2 + 3k + 1$
 $3k^2 - 9k + 1 \geq 0$, true for $k \geq 7$.

$\therefore$ True for $n=k+1$

(k) $z^5 = 1$
 $\therefore z^5 - 1 = 0 \quad , \quad z \neq 1$

(i) $(z+2^{-1}) + (z+z^{-1}) - 1 = 0$

LHS = $z^5 + z + z^{-1} + z + z^{-1} - 1$
 $= z^5 + z^4 + z^3 + z^2 + z + 1 - 1$
 $= z^5 + z^4 + z^3 + z^2 + z + 1 - 1$
 $= \frac{(z-1)(z^5 + z^4 + z^3 + z^2 + z + 1)}{z-1} \quad (A \neq 1)$
 $= \frac{z^5 - 1}{z-1} \quad \text{But } z^5 = 1$
 $= 0 = \text{RHS} \quad \text{As required.}$

or equivalent

(M) $z = e^{i\theta} = \cos\theta + i\sin\theta$
 $z^{-1} = (e^{i\theta})^{-1}$
 $= e^{-i\theta}$
 $= \cos(-\theta) + i\sin(-\theta)$ *cosine = even function*
 $= \cos\theta - i\sin\theta$ *sine = odd function*
 $\therefore z + z^{-1} = \cos\theta + i\sin\theta + \cos\theta - i\sin\theta$
 $= 2\cos\theta$
 $\therefore \cos\theta = \frac{z + z^{-1}}{2} \quad \text{As required.}$

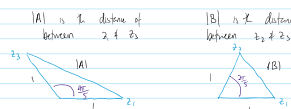
(N) from (i) $(z+z^{-1})^2 + (z+z^{-1}) - 1 = 0$
 let $x = z + z^{-1}$

$x^2 + x - 1 = 0$
 $x = \frac{-1 \pm \sqrt{1+4}}{2}$
 $= \frac{-1 \pm \sqrt{5}}{2}$

z can be any complex root of $z^5 = 1$ *$\cos\theta = \cos\theta + i\sin\theta$*
 let $z = \cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ *$(z^5 = (\cos\frac{2\pi}{5})^5 + i(\sin\frac{2\pi}{5})^5)$*
 $= \cos 2\pi + i\sin 2\pi$ *(By DMPT)*
 $= 1 + i \cdot 0 \quad \therefore z \text{ can be } \cos\frac{2\pi}{5}$

$\therefore 2\cos\frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{2} \quad \text{or} \quad \frac{-1 - \sqrt{5}}{2}$
 $\frac{2\pi}{5}$ is in the first quadrant $\therefore$ true
 $\therefore 2\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{2}$
 $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$

(N) let $A = z_1 - z_2$ $B = z_1 - z_3$ $AB = z_2 - z_3 = \cos\frac{4\pi}{5}$



$|A|^2 = |1 - 2 \times 1 \times \cos\frac{4\pi}{5}|$ $|B|^2 = 2 - 2\cos\frac{2\pi}{5}$

$\frac{|A|^2}{|B|^2} = \frac{2 - 2\cos\frac{4\pi}{5}}{2 - 2\cos\frac{2\pi}{5}}$ *$\cos 2\theta = 2\cos^2\theta - 1$*

$= \frac{2 \left[1 - (2\cos^2\frac{2\pi}{5} - 1) \right]}{2(1 - \cos\frac{2\pi}{5})}$

$= \frac{2 \left[1 - \cos\frac{2\pi}{5} \right]}{2(1 - \cos\frac{2\pi}{5})}$

$= \frac{2 \left[(1 + \cos\frac{2\pi}{5})(1 - \cos\frac{2\pi}{5}) \right]}{2(1 - \cos\frac{2\pi}{5})}$

$= 2 \left[1 + \frac{\sqrt{5}-1}{4} \right]$ *from (iv)*

$= \frac{4 + \sqrt{5} - 1}{2}$

$\left(\frac{|A|}{|B|} \right)^2 = \frac{3 + \sqrt{5}}{2}$ $\left(\frac{1 + \sqrt{5}}{2} \right)^2 = \frac{1 + 5 + 2\sqrt{5}}{4}$

$\therefore \frac{|A|}{|B|} = \sqrt{\frac{3 + \sqrt{5}}{2}}$ $= \frac{1 + \sqrt{5}}{2}$
 $= \frac{1 + \sqrt{5}}{2}$ *As required.*

(b) $r(t) = \cos\frac{\pi}{4}t \hat{i} + \left(\cos\frac{\pi}{4}t + \sin\frac{\pi}{4}t \right) \hat{j} + \sin\frac{\pi}{4}t \hat{k}$

(i) $r(3) = -\frac{1}{\sqrt{2}} \hat{i} + 0 \hat{j} + \frac{1}{\sqrt{2}} \hat{k}$

$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ ✓

(ii) Direction of tangent $\Rightarrow \frac{d}{dt} [\vec{r}(t)]$ @ $t=3$.

$$\frac{d}{dt} \vec{r}(t) = -\frac{\pi}{4} \sin\left(\frac{\pi}{4}t\right) \hat{i} + \left(\frac{\pi}{4} \cos\frac{\pi}{4}t - \frac{\pi}{4} \sin\frac{\pi}{4}t\right) \hat{j} + \frac{\pi}{4} \cos\frac{\pi}{4}t \hat{k} \quad \checkmark$$

$$\therefore \frac{d}{dt} \vec{r}(3) = \frac{\pi}{4} \left[-\frac{1}{\sqrt{2}} \hat{i} - \frac{2}{\sqrt{2}} \hat{j} - \frac{1}{\sqrt{2}} \hat{k} \right]$$

$$= \frac{\pi}{4\sqrt{2}} \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix} \quad \checkmark$$

$$\therefore \text{Vector equation of tangent } L = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \frac{\pi}{4\sqrt{2}} \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \checkmark \text{ when } \lambda \in \mathbb{R}.$$

$$(1) \int_0^4 \frac{1}{\sqrt{1+5x}} \quad \begin{matrix} \text{let } x = t+1 \\ x-1 = t \\ x=0 \Rightarrow t=-1 \\ x=4 \Rightarrow t=3 \end{matrix} \quad \left[\begin{matrix} 2t+9 & u=4 \\ 2t+0 & u=1 \end{matrix} \right] \quad \textcircled{1}$$

$$\therefore \int_1^4 \frac{1}{\sqrt{u}} \cdot 2(u-1) du \quad \frac{d}{du} = 2(u-1)$$

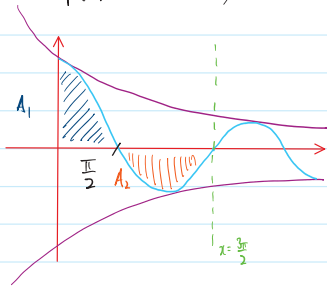
$$= 2 \int_1^4 \left(\frac{u}{\sqrt{u}} - \frac{1}{\sqrt{u}} \right) du$$

$$= 2 \left[\frac{2}{3} u^{3/2} - 2 u^{1/2} \right]_1^4 \quad \textcircled{1}$$

$$= 2 \left[\left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 2 \right) \right] \quad \textcircled{1}$$

$$= \frac{14}{3} \ln 3$$

(a) $f(x) = e^x \cos x$, $x \in [0, \frac{3\pi}{2}]$



$$\text{RTP } \left| \frac{A_1}{A_2} \right| = \frac{e^\pi (e^{\frac{\pi}{2}} + 1)}{e^\pi + 1}$$

$$\int e^{-x} \cos x \, dx \quad \begin{array}{ll} u = e^{-x} & V = \sin x \\ u' = -e^{-x} & V' = \cos x \end{array}$$

$$\therefore I = (e^{-x} \sin x) + \int e^{-x} \sin x \, dx$$

$$\begin{array}{ll} u = e^{-x} & V = -\cos x \\ u' = -e^{-x} & V' = \sin x \end{array}$$

$$= e^{-x} \sin x + \left[-e^{-x} \cos x - \int e^{-x} \cos x \, dx \right]$$

$\Downarrow$
 I

$$\therefore 2I = e^{-x} [\sin x - \cos x]$$

$$I = \frac{e^{-x}}{2} [\sin x - \cos x] \quad (1)$$

Splitting @ $x = \frac{\pi}{2}$ (1)

switched as A_2 should be positive.

$$\therefore A_1 = \left[\frac{e^{-x}}{2} (\sin x - \cos x) \right]_0^{\frac{\pi}{2}}$$

$$A_2 = \left[\frac{e^{-x}}{2} (\cos x - \sin x) \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$$

$$= \left[\frac{e^{-\frac{\pi}{2}}}{2} (1 - 0) - \frac{1}{2} (0 - 1) \right] = \left(\frac{e^{-\frac{3\pi}{2}}}{2} (0 + 1) - \frac{e^{-\frac{\pi}{2}}}{2} (0 - 1) \right)$$

$$= \frac{1}{2} [e^{-\frac{\pi}{2}} + 1]$$

$$= \frac{1}{2} (e^{-\frac{3\pi}{2}} + e^{-\frac{\pi}{2}})$$

$$\frac{A_1}{A_2} = \frac{\frac{1}{2} (e^{-\frac{\pi}{2}} + 1)}{\frac{1}{2} (e^{-\frac{3\pi}{2}} + e^{-\frac{\pi}{2}})}$$

$$= \frac{e^{-\frac{\pi}{2}} (1 + e^{\frac{\pi}{2}})}{e^{-\frac{3\pi}{2}} (1 + e^\pi)}$$

$$= e^{-\frac{\pi}{2} + \frac{3\pi}{2}} \left(\frac{1 + e^{\frac{\pi}{2}}}{1 + e^\pi} \right)$$

$$\begin{aligned}
 A_2 &= \frac{1}{2} (e^{-\frac{3\pi}{2}} + e^{-\frac{\pi}{2}}) \\
 &= \frac{e^{-\frac{\pi}{2}} (1 + e^{\frac{\pi}{2}})}{e^{-\frac{3\pi}{2}} (1 + e^{\pi})} \\
 &= e^{-\frac{\pi}{2} + \frac{3\pi}{2}} \left(\frac{1 + e^{\frac{\pi}{2}}}{1 + e^{\pi}} \right) \\
 &= \frac{e^{\pi} (1 + e^{\frac{\pi}{2}})}{1 + e^{\pi}} \quad A_3 \text{ required}
 \end{aligned}$$

①

(b) $T_n = 6T_{n-1} - 9T_{n-2}$ for $n \geq 2$ $T_0 = 1, T_1 = 6$

RTP $T_n = 3^n + n \times 3^n$

① Show true for $n=0$ & $n=1$

$$T_0 = 3^0 + 0 \times 3^0$$

$$= 1 \quad \therefore \text{true for } n=0$$

$$T_1 = 3^1 + 1 \times 3^1$$

$$= 6$$

$$\therefore \text{true for } n=1$$

② Assume true for $n=k$ & $n=k-1$ (k integer, $k \geq 1$)

$$T_k = 3^k + k \times 3^k$$

$$T_{k-1} = 3^{k-1} + (k-1) \times 3^{k-1}$$

③ Prove true for $n=k+1$

RTP $T_{k+1} = 3^{k+1} + (k+1) 3^{k+1}$

$$T_{k+1} = 6T_k - 9T_{k-1}$$

$$= 6(3^k + k \times 3^k) - 9(3^{k-1} + (k-1)3^{k-1})$$

$$= 6 \times 3^k + 6k \times 3^k - 9 \times 3^{k-1} - 9(k-1)3^{k-1}$$

$$= 6 \times 3^k + 6k \times 3^k - 3 \times 3^k - 3(k-1)3^k$$

$$= 3 \times 3^k + (6k - 3k + 3) 3^k$$

$$= 3^{k+1} + (3k+3) 3^k$$

from Assumption. ①

$$\begin{aligned}
 &= 3^{k+1} + (3k+3)3^k \\
 &= 3^{k+1} + 3(k+1)3^k \\
 &= 3^{k+1} + (k+1)3^{k+1} \quad \text{As required}
 \end{aligned}$$

True for $n=k+1$

Proven by Mathematical Induction.

(c) let $I_n = \int \frac{\cos nx}{\sin x} dx$

(i) RTP $\cos(n-2)x - \cos nx = 2 \sin((n-1)x) \sin x$

from reference sheet.

$$2 \sin A \sin B = [\cos(A-B) - \cos(A+B)] \quad (1)$$

let $A = (n-1)x$ $B = x$

$$\begin{aligned}
 2 \sin[(n-1)x] \sin x &= \cos[(n-1)x - x] - \cos[(n-1)x + x] \\
 &= \cos(n-2)x - \cos nx \quad \text{As required.} \quad (2)
 \end{aligned}$$

(ii) $I_n = \int \frac{\cos nx}{\sin x} dx$

$$I_{n-2} = \int \frac{\cos(n-2)x}{\sin x} dx$$

$$I_n - I_{n-2} = \int \frac{\cos nx - \cos(n-2)x}{\sin x} dx$$

$$= \int \frac{- (2 \sin(n-1)x) \cancel{\sin x}}{\cancel{\sin x}} dx \quad \text{from (i)}$$

$$= - \int 2 \sin(n-1)x dx$$

$$= \frac{\cos(n-1)x}{n-1} + C$$

$$= \frac{2 \cos(n-1)x}{n-1} + C \quad \text{As required.}$$

$n-1$

✓

(iii) $\int_0^{\frac{\pi}{3}} \frac{\cos 2x - \cos 6x}{\sin x} dx$

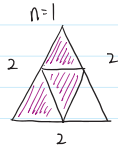
$$= I_2 - I_6 + I_4 - I_4 \quad (1)$$

$$= -[I_6 - I_4] - [I_4 - I_2]$$
$$= - \left[\left[\frac{2 \cos(5)x}{5} \right] + \left[\frac{2 \cos(3)x}{3} \right] \right]_0^{\frac{\pi}{3}} \quad (1)$$

$$= - \left[\left(\frac{2}{5} \times \frac{1}{2} + \frac{2}{3} \times -1 \right) - \left(\frac{2}{5} + \frac{2}{3} \right) \right]$$

$$= \frac{23}{15} \text{ units}^2 \quad (1)$$

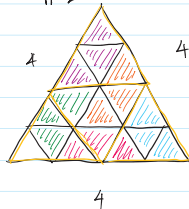
When



- 1 piece needed
- 1 empty space

$$\text{Triangle Area} = 4$$

$\therefore$ true for $n=1$

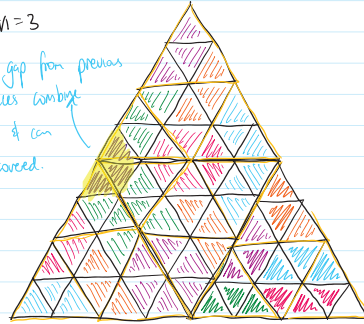
 $n=2$ 

- 5 pieces needed
- 1 empty space

$$\text{Triangle area} = 4 \times 4 = 16$$

 $n=3$

gap from previous
pieces combine
here & can
be covered.



- 21 pieces needed
- 1 empty space

$$\text{Triangle area} = 64$$

$$n=4, \text{ pieces needed} = 4 \times 21 + 1 = 85$$

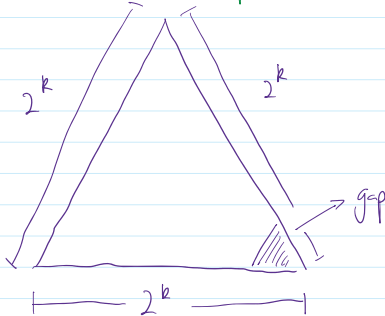
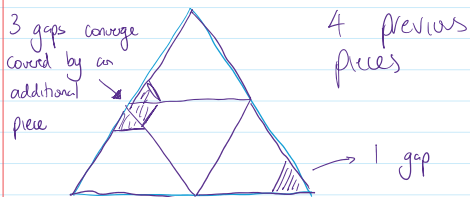
$$\text{Triangle Area} = 256$$

Each increase in n

requires 4 of the previous triangle.

3 can rotate to create a new piece

$\therefore$ if area is true the gap always in a corner.

Assume true for $n=k$ Prove true for $n=k+1$ 

$\therefore$ True for $n=k+1$

Thus proven by Mathematical induction.

① True for $n=1$ ② Displaying the k th piece③ Forming the $(k+1)$ th piece from 4 previous pieces.

$$(b) \quad S = \frac{a}{1-r}$$

$$(i) \quad a=1 \quad r = \frac{1}{2}e^{i\theta} \quad \textcircled{1}$$

$$\therefore S = \frac{1}{1 - \frac{1}{2}e^{i\theta}}$$

$$= \frac{1}{\frac{2 - e^{i\theta}}{2}}$$

$$= \frac{2}{2 - e^{i\theta}} \quad \text{As required.} \quad \textcircled{1}$$

$$\therefore A_n e^{i\theta} = r^n A + i^n A$$

$$\frac{1}{2-e^{i\theta}}$$

(ii) As $e^{i\theta} = \cos\theta + i\sin\theta$

$$\operatorname{Im}(S) = \frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \dots \quad (1)$$

$$\therefore \frac{1}{2}\sin\theta + \frac{1}{4}\sin 2\theta + \frac{1}{8}\sin 3\theta + \dots = \operatorname{Im}\left(\frac{2}{2-e^{i\theta}}\right)$$

$$\begin{aligned} \operatorname{Re}(S) &= \operatorname{Im}\left(\frac{2}{2-\cos\theta-i\sin\theta} \times \frac{2-\cos\theta+i\sin\theta}{2-\cos\theta+i\sin\theta}\right) \\ &= \operatorname{Im}\left[\frac{2(2-\cos\theta) + 2i\sin\theta}{(2-\cos\theta)^2 + \sin^2\theta}\right] \\ &= \frac{2\sin\theta}{4 - 4\cos\theta + \cos^2\theta + \sin^2\theta} \\ &= \frac{2\sin\theta}{5 - 4\cos\theta} \end{aligned} \quad (1)$$

(iii) For the sum to be purely imaginary then

$$\operatorname{Re}(S) = 0$$

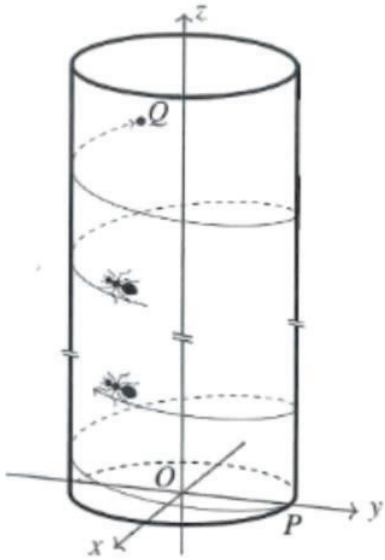
$$\operatorname{Re}(S) = \frac{2(2-\cos\theta)}{5-4\cos\theta} \quad (1)$$

$$\operatorname{Re}(S) = 0 \quad \text{When} \quad 2-\cos\theta = 0$$

$$-1 \leq \cos\theta \leq 1 \quad \therefore \quad 2-\cos\theta \neq 0$$

$\therefore$ Sum can never be purely imaginary.

(c)



$$\underline{r}(t) = \left(7 \sin \frac{\pi}{32} t\right) \underline{i} + \left(7 \cos \frac{\pi}{32} t\right) \underline{j} + \frac{t}{15} \underline{k}$$

$$(i) \quad |OQ| = |\underline{r}(t)|$$

$$|\underline{r}(t)|^2 = 49 \left[\sin^2 \frac{\pi}{32} t + \cos^2 \frac{\pi}{32} t \right] + \frac{t^2}{225}$$

$$\therefore 49 + \frac{t^2}{225} = 25^2$$

$$t^2 = 576 \times 225$$

$$t = 360 \quad (t > 0 \text{ as time is positive})$$

$$\text{Position of } Q = \underline{r}(360)$$

$$= 7 \sin \frac{360\pi}{32} \underline{i} + 7 \cos \frac{360\pi}{32} \underline{j} + \frac{360}{15} \underline{k}$$

$$= \left(-\frac{7}{\sqrt{2}}, -\frac{7}{\sqrt{2}}, 24 \right)$$

(ii) If $\underline{r}$ component of $\underline{r}(t) = 7$

$$7 = 7 \sin \frac{\pi t}{32} \quad (1)$$

$$\sin \frac{\pi t}{32} = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} \dots$$

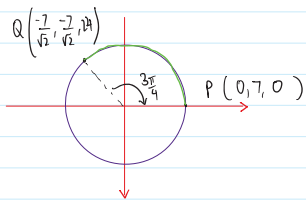
$$\therefore t = 16, 80, 144, 208, 272, 336, \boxed{400} \dots$$

$> 360 \therefore$ Ant has stopped moving

$\therefore$ Crosses $x=7$ 6 times (1)

(iii) From $Q \rightarrow P$ the ant walks along the curved edge of the cylinder

From above.

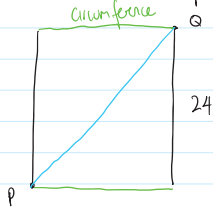


That is the ant's trip
can be considered in
2 parts

- Along the circumference of the circle in the x/y plane
- 24 units down in the z plane.

① Any working towards the components of the trip

This can be represented as a rectangle



$$\begin{aligned}\text{Circumference} &= r\theta \\ &= 7 \times \frac{3\pi}{4} \\ &= \frac{21\pi}{4}\end{aligned}$$

$$\therefore \text{Shortest path} = \sqrt{24^2 + \left(\frac{21\pi}{4}\right)^2}$$

$$\div 29.1 \text{ cm. } \textcircled{1}$$